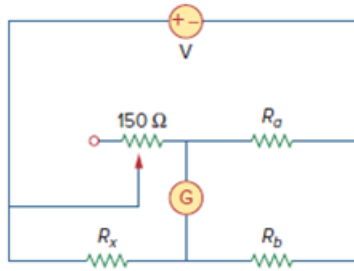


Problem

(a) In the Wheatstone bridge circuit of Fig. 4.147 select the values of R_a and R_b such that the bridge can measure R_x in the range of 0–25 Ω .

(b) Repeat for the range of 0–250 Ω .



Step-by-step solution

Step 1 of 2

Refer to Figure 4.147 in the text book.

(a)

The value of the resistance R_a should be equal to the value of $150\ \Omega$ for the bridge to be balanced that is, to set the value of the galvanometer to zero.

Similarly, the value of the resistance R_b should be equal to the highest value of R_x for the bridge to be balanced that is, to set the value of the galvanometer to zero. Therefore, the value of the resistance R_b is, $25\ \Omega$.

Step 2 of 2

(b)

The value of the resistance R_a should be equal to the value of $150\ \Omega$ for the bridge to be balanced that is, to set the value of the galvanometer to zero.

Also the value of the resistance R_b should be equal to the highest value of R_x for the bridge to be balanced that is, to set the value of the galvanometer to zero. Therefore, the value of the resistance R_b is, $250\ \Omega$.